

Module 1 – Introduction to Networking

Basics of Computer Networks

- A network is an interconnection of devices for resource and data sharing.
- It improves communication, storage, and centralized control.
- LAN / MAN / WAN / PAN differ in range and cost.
- Used in offices, education, banking, and research.
- Enhances collaboration and efficiency.

Components of a Computer Network

- NIC connects a computer to the network and provides a MAC address.
- Switches, routers, and gateways control data flow and routing.
- Transmission media may be wired (copper/fiber) or wireless (radio).
- Each device works at a specific OSI layer.
- Together they enable connectivity and control.

Network Types & Topologies

- Peer-to-peer and client-server define network roles.
- Topologies: Bus, Star, Ring, Mesh, Tree, Hybrid.
- Star offers reliability; bus is cost-effective.
- Topology affects performance and fault-tolerance.
- Chosen based on scale and budget.

Networking Devices

- Repeater amplifies signals to extend distance.
- Hub broadcasts data to all ports; switch forwards selectively.
- Router connects different networks; gateway translates protocols.
- Access point links wireless clients to LAN.
- Modem converts digital–analog signals for ISP access.

Basic Communication System

- Involves sender, receiver, medium, encoder/decoder, and message.
- Simplex, half-duplex, and full-duplex define data direction.
- Protocols standardize format and timing (e.g., TCP/IP).
- Noise and attenuation degrade signals over distance.
- Feedback ensures reliable exchange.

Transmission and Signal Concepts

- Digital signals use discrete bits; analog are continuous waves.
- Bandwidth controls channel data capacity.
- Noise and interference affect throughput.
- Encoding and modulation maintain synchronization.
- Amplifiers restore weakened signals.

Switching and Multiplexing

- Circuit switching creates a dedicated path for voice calls.
- Packet switching divides data into packets for Internet use.
- Message switching stores and forwards entire messages.
- FDM, TDM, WDM share a medium by frequency, time, or wavelength.
- Used to optimize channel usage and efficiency.

OSI Reference Model

- Defines seven layers for standard communication.
- Each layer offers services to the layer above.
- Physical → Application layers handle signals to user apps.
- Encapsulation adds headers; decapsulation removes them.
- Facilitates interoperability and troubleshooting.

Module 2 – Data Link Layer

Introduction

- Connects Network and Physical layers for reliable frames.
- Responsible for framing, error detection, and flow control.
- Operates between nodes on the same link.
- Ensures error-free data transfer to Network layer.
- Divided into LLC and MAC sublayers.

Error Detection and Correction

- Parity, Checksum, and CRC identify bit errors.
- Hamming Code can locate and correct single-bit errors.
- Redundancy bits added to data for verification.
- Detected frames are retransmitted.
- Improves reliability on noisy links.

Data Link Protocols

- Stop-and-Wait ARQ uses ACK before next frame.
- Go-Back-N transmits multiple frames before ACK.
- Sliding Window controls sequence numbers and flow.
- Protocols balance efficiency and reliability.
- Used in LAN and WAN data exchange.

MAC & LLC Sublayers

- MAC handles addressing and channel access.
- LLC provides interface to upper layers.
- IEEE 802 standards define operations.
- MAC uses CSMA/CD or CSMA/CA.
- LLC adds flow and error control info.

Data Link Control

- Ensures ordered and error-free delivery.
- Uses ACK/NACK frames and timers.
- Framing adds headers and trailers for synchronization.
- Flow control prevents buffer overflow.
- Implemented in PPP and HDLC.

Channelization Techniques

- FDMA divides frequency band among users.
- TDMA assigns time slots.
- CDMA uses unique codes per user.
- Improve bandwidth utilization and security.
- Used in cellular and satellite systems.

Wired LANs

- Ethernet uses CSMA/CD access method.
- Frames contain source, destination, and CRC.
- Fast/Gigabit Ethernet increase speed using switches.
- 10 Gig Ethernet for backbones.
- Supports low latency and high throughput.

Wireless LANs

- IEEE 802.11 defines Wi-Fi standards.
- Operate in 2.4 GHz or 5 GHz bands.
- Infrastructure and Ad-hoc modes available.
- Use WPA/WPA2 security mechanisms.
- SSID identifies the wireless network.

Module 3 – Introduction to Network Layer

Overview of Network Layer

- Responsible for routing and logical addressing.
- Transfers packets between networks.
- Performs fragmentation and reassembly.
- Supports connectionless communication.
- Ensures efficient path selection.

Design Issues

- Focuses on packet routing and address scheme.
- Handles inter-network communication.
- Provides error handling and congestion control.
- Balances efficiency and reliability.
- Manages connection-oriented and connectionless services.

Routing Algorithms

- Static routing uses manual entries.
- Dynamic routing adapts to network changes.
- Distance Vector uses hop count as metric.
- Link State uses topology updates.
- Hybrid approach combines both.

Congestion Control

- Occurs when traffic exceeds capacity.
- Open-loop prevents before occurrence.
- Closed-loop detects and reacts to congestion.
- Queuing and traffic shaping help control.
- Ensures fair resource usage.

Quality of Service (QoS)

- Measures bandwidth, delay, jitter, and loss.
- Prioritizes real-time traffic.
- Resource reservation improves performance.
- Implemented in VoIP and streaming.
- Supports service level agreements.

Network Layer Services

- Connectionless service uses datagrams.
- Connection-oriented sets up virtual paths.
- Provides best-effort delivery.
- Handles fragmentation and reassembly.
- Manages logical addressing.

Packet Switching & Forwarding

- Datagram networks route each packet independently.
- Virtual circuits establish logical connections.
- Forwarding uses routing tables.
- Routers decide next hop based on IP.
- Increases network efficiency.

IPv4 Addressing

- Divided into classes A to E.
- Subnetting creates smaller logical networks.
- Supernetting combines multiple networks.
- CIDR provides flexible prefix notation.
- IP addresses are unique identifiers.

Protocols

- IP handles packet delivery and routing.
- ICMPv4 reports errors and diagnostic messages.
- ARP resolves IP to MAC address.
- Mobile IP supports roaming devices.
- Unicast routing delivers packets to single hosts.

Module 4 – Transport Layer & Application Layer

Transport Layer Fundamentals

- Ensures end-to-end communication.
- Provides multiplexing and demultiplexing of ports.
- Handles segmentation and reassembly.
- Offers flow and error control.
- Delivers data reliably to applications.

Types of Transport

- Connection-oriented (TCP) sets up sessions.
- Connectionless (UDP) sends independent packets.
- Reliable uses ACK and retransmission.
- Unreliable trades speed for simplicity.
- Choice depends on application needs.

Transport Protocols

- TCP uses three-way handshake for connection.
- Provides sequencing and error recovery.
- UDP is lightweight and fast.
- TCP ensures ordered delivery; UDP does not.
- Used in web browsing, streaming, and gaming.

Congestion Control (TCP)

- Prevents packet loss and network collapse.
- AIMD adjusts window size dynamically.
- Slow start increases rate gradually.
- Fast retransmit handles missing ACKs.
- Maintains fairness among connections.

Application Layer Overview

- Topmost layer providing user interfaces.
- Supports network applications like email and FTP.
- Implements client-server model.
- Uses protocols defined by services.
- Translates user requests to network operations.

Email Protocols

- SMTP sends messages between servers.
- POP3 and IMAP retrieve mail to clients.
- MIME adds attachments and multimedia.
- Provide authentication and delivery status.
- Form the basis of Internet mail system.

File Transfer and Remote Access

- FTP transfers files using TCP connection.
- TELNET enables remote login to servers.
- Both use client-server architecture.
- Support authentication and command operations.
- Replaced partly by SFTP and SSH.

Configuration and Management Protocols

- DHCP assigns IP addresses dynamically.
- Reduces manual configuration errors.
- SNMP monitors network devices via agents.
- Uses MIB database for management information.
- Crucial for enterprise networks.

Module 5 – Client-Side Scripting with JavaScript & DOM

Introduction to JavaScript

- Interpreted scripting language for web pages.
- Adds interactivity and logic to HTML.
- Supports variables, operators, and control structures.
- Runs in browsers without compilation.
- Widely used in frontend development.

DOM (Document Object Model)

- Represents HTML as a tree of objects.
- Allows access and modification of elements.
- Supports event handling (mouse, keyboard).
- Dynamic updates without page reload (through JS).
- Enables interactive and responsive UI.

JavaScript in Web Pages

- Scripts embedded using `<script>` tags.
- Can modify content, style, and structure runtime.
- Handles forms and validations.
- Enhances user experience via animations.
- Cross-browser testing is important.

Window and Document Objects

- Window is global browser object containing document.
- Document represents the current HTML page.
- Provide methods like `alert()`, `getElementById()`.
- Used for pop-ups and DOM access.
- Control browser view and content.

Case Study (JS + DOM)

- Combines HTML, CSS, and JavaScript for UI.
- Validates form inputs dynamically.
- Displays results without page reload.
- Improves usability and performance.
- Forms basis for modern web apps.

XML Basics

- Extensible Markup Language for data representation.
- Uses tags defined by users for structure.
- Supports nested elements and attributes.
- Platform-independent data exchange format.
- Used in config files and web services.

Module 6 – XML and Server-Side Scripting with PHP / Servlets

Markup Languages (XHTML)

- Stricter version of HTML using XML syntax.
- Requires proper tag nesting and lowercase names.

- Improves browser compatibility and validation.
- Supports separation of content and style.
- Used for well-structured web documents.

Servlet Architecture

- Server-side Java components handling HTTP requests.
- Run inside a web container like Tomcat.
- Use Request and Response objects for I/O.
- Platform-independent and scalable.
- Replaces CGI for efficiency.

Servlet Life Cycle

- `init()` executed once for initialization.
- `service()` handles GET/POST requests.
- `destroy()` called before termination.
- Managed by the Servlet container.
- Ensures reusability and resource management.

Generating Dynamic Content

- Servlets embed HTML in Java code to respond to clients.
- Use `PrintWriter` to output HTML data.
- Can fetch form data and display results.
- Integrate with JDBC for database content.
- Provide real-time page generation.

Parameter Data Handling

- `request.getParameter()` retrieves form inputs.
- Supports GET and POST methods.
- Used for login and registration forms.
- Validates data before processing.
- Improves interaction between client and server.

Session Management

- Maintains state across multiple requests.
- Uses Cookies, Hidden Fields, URL Rewriting, `HttpSession`.
- Tracks user identity and preferences.
- Essential for shopping carts and logins.
- Secured through timeouts and HTTPS.

Additional Servlet Capabilities

- Support for JDBC database connections.
- Handle concurrency using threads.
- Manage files and server resources.
- Enable filters and listeners for modular design.
- Integrate with JSP for advanced applications.

Case Study (Dynamic Web App)

- A simple client-server web form using Servlets.
- Collects user data and stores in database.
- Displays results dynamically using HTML output.
- Demonstrates session and cookie usage.
- Forms foundation for enterprise web systems.

OSI Layer-wise Protocol List (7 Each)

1. Physical Layer (Layer 1)

Function: Transmission of raw bits over a physical medium.

Protocols / Standards:

1. Ethernet (IEEE 802.3 – physical specs)
2. USB (Universal Serial Bus)
3. Bluetooth (IEEE 802.15.1)
4. DSL (Digital Subscriber Line)
5. SONET / SDH (Synchronous Optical Networking)
6. Wi-Fi Physical Layer (IEEE 802.11 PHY)
7. RS-232 (Serial Communication Standard)

2. Data Link Layer (Layer 2)

Function: Node-to-node data transfer, framing, and error control.

Protocols / Standards:

1. Ethernet (IEEE 802.3)
2. PPP (Point-to-Point Protocol)
3. HDLC (High-Level Data Link Control)
4. Frame Relay
5. ATM (Asynchronous Transfer Mode – Layer 2 part)
6. Token Ring (IEEE 802.5)
7. MAC (Media Access Control – part of IEEE 802 family)

3. Network Layer (Layer 3)

Function: Logical addressing, routing, and path determination.

Protocols / Standards:

1. IP (Internet Protocol – IPv4 / IPv6)
2. ICMP (Internet Control Message Protocol)
3. IGMP (Internet Group Management Protocol)
4. IPsec (Internet Protocol Security)
5. ARP (Address Resolution Protocol)
6. RIP (Routing Information Protocol)
7. OSPF (Open Shortest Path First)

4. Transport Layer (Layer 4)

Function: End-to-end communication, segmentation, flow and error control.

Protocols / Standards:

1. TCP (Transmission Control Protocol)
2. UDP (User Datagram Protocol)
3. SCTP (Stream Control Transmission Protocol)
4. DCCP (Datagram Congestion Control Protocol)
5. SPX (Sequenced Packet Exchange)
6. RTP (Real-Time Transport Protocol)
7. RSVP (Resource Reservation Protocol)

5. Session Layer (Layer 5)

Function: Establish, manage, and terminate sessions between systems.

Protocols / Standards:

1. NetBIOS (Network Basic Input/Output System)
2. RPC (Remote Procedure Call)
3. PPTP (Point-to-Point Tunneling Protocol)
4. SOCKS (Session proxy protocol)
5. NFS (Network File System – session management)
6. AppleTalk Session Protocol (ASP)
7. Session Control Protocol (SCP)

6. Presentation Layer (Layer 6)

Function: Data translation, encryption, compression, and formatting.

Protocols / Standards:

1. SSL / TLS (Secure Sockets Layer / Transport Layer Security)
2. MIME (Multipurpose Internet Mail Extensions)
3. JPEG / PNG (Image encoding standards)
4. ASCII / EBCDIC (Character encoding formats)
5. XDR (External Data Representation)
6. MPEG (Multimedia Encoding Standard)
7. XML / JSON (Data representation and structure)

7. Application Layer (Layer 7)

Function: User interface and network application services.

Protocols / Standards:

1. HTTP / HTTPS (Web Communication)
2. FTP (File Transfer Protocol)
3. SMTP (Simple Mail Transfer Protocol)
4. POP3 (Post Office Protocol v3)
5. IMAP (Internet Message Access Protocol)
6. DNS (Domain Name System)
7. SNMP (Simple Network Management Protocol)